

HEAT AND FLUID FLOW IN WELDS

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Abstract

The current state and future directions of numerical modeling of heat and fluid flow in welds are presented. Weld pool models have evolved rapidly in the past decade and full 3D analyses are now possible for short welds. In two dimensions, modeling the formation of defects has begun. Outside of the weld pool, computational heat flow analyses are now able to analyze welds of medium length with useful resolution and accuracy. Prescribed flux and power density functions or prescribed temperature functions are used to model the heating effect of the arc. Kinematic models have evolved from 2D cross-sectional to 2D in-plane to full 3D and shell-3D composite models. The coupling of heat flow models to the evolution of microstructure and thermal stress analysis has resolved the effect of transformations on the residual stress in high strength steels. The paper closes with a discussion of future opportunities and challenges.

Key words: Computational Weld mechanics, Weld pool, Heat transfer, Arc welding.

1. The Weld Pool

Our understanding of the physics of the weld pool prior to 1980 is clearly explained in a beautiful book written by J. F. Lancaster [1]. The basic physics of surface tension in fusion welding is discussed in detail in the four part paper by Matsunawa [2]. Heiple and Roper's experimental studies strongly influenced weld pool research in the past decade [3]. Specific weld pool models are described in references [4-12].

Prior to 1980, the primary mechanism for heat transfer in weld pools was known to be convection, i.e., heat is transported by the motion of the liquid and not by diffusion. Although the basic forces acting on the weld pool were known to be due to buoyancy, electromagnetic, Marangoni and arc pressure, the relative importance of these forces was not well understood. This question was resolved by the numerical models that began to appear in the 1980s [4-12]. These models not only demonstrated that the effect of buoyancy, electromagnetic, Marangoni and arc pressure forces on the weld pool velocity, pressure and temperature could be computed but also demonstrated that numerical modeling was a powerful tool for understanding the weld pool.

It can be argued that Zacharia, Ersalan, et al, presented the most comprehensive analysis of the weld pool performed to date [10,11]. Their model was fully 3D and transient. The weld pool-arc and liquid-solid interfaces were free boundaries. Filler metal was added to the weld pool from the arc. Computational costs limited the arc travel to a short distance. MacLay and Carey used a finite element formulation for an axisymmetric weld pool with both flat and depressed weld pool surfaces. As more distorted weld pools are studied, the use of FEM is likely to become more important. Kou studied the effect of the normal and shear stress induced on the weld pool surface due to surface tension in a deformed weld pool [7]. Mazumder considered the numerical requirements in some detail [6]. More recently, Szekely [9] and Matsunawa [9] have both studied the effect of sloshing in the weld pool.

Although progress in modeling the weld pool is impressive, the capability of current models to predict weld pool behavior is limited by several factors. The data required include the heat flux, current density distribution and pressure velocity field from the arc; the surface tension as a function of composition and temperature of the weld pool surface; the viscosity, density and vapor pressure of the liquid metal in the weld pool. Those data are difficult to obtain, the accuracy is modest and they cannot yet be predicted for most welding procedures. The computing costs are high. These factors have forced most analyses to be 2D and often stationary. The arc interacts with the weld pool shape and the weld pool shape affects the arc. Therefore it will be important to study not only the weld pool but the weld pool and arc as a single coupled system. This is particularly important for weld pools with deep depressions. Szekely has made a useful start on coupling the arc and weld pool [9]. The primary contribution of weld pool models to date is in understanding the behavior of weld pools and the importance of various factors. Although it will be some time before weld pool models realize their ultimate dream; to predict the weld pool shape for a given welding procedure, there is little doubt that to truly understand, control or predict welds and welding processes, we must model weld pools accurately and efficiently.

Remark: Because of limited space only the most critical or most recent references are cited to provide the reader with an entry into the literature. We apologize to authors of references that have been omitted.

2. Transient Heat Affected Zone

2.1 Background

The classical analysis of heat flow in welding was pioneered by Rykalin and Rosenthal [13, 14]. Rykalin's Houdremont lecture of 1972 and the review paper by Myers et al [15] summarize the research from this classical viewpoint. This paper focuses on numerical methods which have dominated the study of heat flow in welds since Westby's pioneering analysis of 1965 [16]. The major research issues are; i) modeling the distribution of heat input from the arc; ii) the kinematic model - 2D cross-sectional, 2D in-plane, 3D or 3D-shell composite model; iii) the design of FEM meshes; iv) the choice of solver and time integration scheme; v) the computational cost and accuracy obtained; vi) the study of errors; vii) the possibilities of obtaining optimal solutions; viii) coupling heat transfer to microstructure, thermal stress analysis including transformation plasticity; ix) unsolved problems including the problem of multi-pass welds.

2.2 Theory

The conservation of energy is the fundamental principle that underlies the analysis of heat flow. It is usually expressed in the form of a partial differential equation which can be found in any text on heat transfer. A finite difference approximation to the derivatives leads to the finite difference method. The error depends on the difference between the finite difference approximation and the derivative. However, finite element analysis, FEM, is based on an integral formulation which minimizes the error in a least square sense, i.e., the error is made orthogonal to the test function space. The complete statement of a well posed heat flow problem requires the domain in space (geometry), the boundary conditions, initial conditions and the thermal properties be specified and consistent.